



Emotional autonomy, psychosocial adjustment and parenting: interactions, moderating and mediating effects

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Following inconsistent results on the developmental outcomes of emotional autonomy, this study examined the consequences of emotional and behavioural autonomy for different aspects of psychosocial adjustment in the context of the parenting process as perceived by the adolescent. Measures of emotional autonomy, behavioural autonomy, perceived parenting and various aspects of psychosocial adjustment were completed by a sample of 558 adolescent boys and girls (aged 12–17 years) from the Dutch-speaking part of Belgium. A continuous measure of authoritativeness was constructed for this study. Correlations revealed that authoritativeness was associated with a positive pattern of adjustment, while autonomy was associated with a negative pattern of adjustment, except for self-reliance. Analyses indicated that authoritativeness and autonomy did not interact to produce mean differences in adjustment. Further, hierarchical multiple regression analyses showed that it was only for self-reliance that the interaction between emotional autonomy and authoritativeness explained significantly more variance than a model that comprised only the main effects. Finally, path analysis clarified the difference between emotional and behavioural autonomy, in that emotional autonomy predicted only higher levels of internal distress, while behavioural autonomy predicted only lower school grades and higher levels of deviant behaviour, at least when authoritativeness was partialled out. These results reveal that it is useful to study autonomy and its developmental outcomes in the broader family context, but they also reveal the limitations of such an approach.

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Introduction

The present study deals with emotional autonomy and its developmental outcomes. Emotional autonomy occupies a central place in psychoanalytic and neoanalytic theories about adolescence. Blos (1962, 1979) claimed that one of the results of the second individuation process during adolescence is emotional autonomy. Rather than detach from their parents, adolescents relinquish childish conceptualizations of their parents and dependencies on them. Part of this process is the de-idealization of parents or disengagement from the infantile parental objects. On the one hand, this disengagement from the infantile object results in ego maturation (Blos, 1979). On the other hand, the development of emotional autonomy involves loss and consequently distress (Hill and Holmbeck, 1986).

Steinberg and Silverberg (1986) operationalized the construct of emotional autonomy in a study of young adolescents. They developed the Emotional Autonomy Scale (EAS), a 20-item measure, assessing four components of emotional autonomy: non-dependency, individuation, parental de-idealization and perceiving one's parents as people. Items were

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written to be consistent with the neoanalytic perspective on emotional autonomy during adolescence (Blos, 1979), which de-emphasizes the storm and stress of adolescent detachment or rebellion and focuses, instead, on the more peaceful process of individuation.

Steinberg and Silverberg (1986) stated that the development of emotional autonomy is adaptive in the long run, because emotional distance from parents is needed for adolescents to begin to rely on their own internal resources. Since then, one of the most lively debates in the recent literature on autonomy at adolescence is whether emotional autonomy from parents really serves this adaptive function (Silverberg and Gondoli, 1996). Research about the consequences of emotional autonomy in early adolescence, as measured by the EAS, has yielded inconsistent results. On the one hand, some authors have suggested that emotional autonomy has positive developmental outcomes. Chen and Dornbusch (1998) examined the differential correlates of various aspects of emotional autonomy. High de-idealization and non-dependency, the alleged core aspects of emotional autonomy, were associated with positive outcomes such as better school grades.

On the other hand, Ryan and Lynch (1989) reported that high emotional autonomy was related to greater feelings of insecurity with parents and lower feelings of lovability, both in adolescent and young adult samples. The authors concluded that those youngsters who experienced high levels of emotional autonomy, as measured by the EAS, may be experiencing feelings of detachment. Several authors lent support to this conclusion by reporting negative associations between emotional autonomy or detachment and a series of psychosocial outcome variables (Turner *et al.*, 1991, 1993; Papini and Roggman, 1992; Feiring and Lewis, 1993).

Following these contradictory results, Lamborn and Steinberg (1993) proposed that the meaning of emotional autonomy (an individual characteristic of the adolescent) and its consequences for psychosocial adjustment could only be understood if studied in context. The meaning of adolescent emotional autonomy may differ depending on the quality of the parent-adolescent relationship. More specifically, the authors demonstrated that emotionally autonomous adolescents who also perceived their parents as being unsupportive (labelled as detached adolescents), showed a negative pattern of adjustment and competence. However, emotionally autonomous adolescents who perceived high levels of parental support (individuated adolescents) showed a positive pattern of adjustment and competence. In some aspects (e.g. academic competence) these youngsters were better adjusted than adolescents who reported high levels of parental support and low levels of emotional autonomy (connected adolescents). In a similar study using cluster analysis, Delaney (1996) identified a cluster of individuated adolescents who reported high levels of closeness and moderate to high levels of emotional autonomy. These individuated adolescents not only formed the large majority of the sample, but also were less anxious, less depressed and had higher self-esteem than did adolescents who had detached relationships with their parents (low levels of closeness and high levels of emotional autonomy).

However, applying the same contextual-moderator framework, Fuhrman and Holmbeck (1995) showed that it is only under conditions of family stress (low maternal warmth, low family cohesion, parent-adolescent conflict, or non-intact family structures) that feelings of emotional autonomy are adaptive. They reasoned that under these conditions, emotional autonomy serves a protective function and facilitates development. Under less stressful and more supportive conditions, high emotional autonomy would be maladaptive. McClanahan and Holmbeck (1992) showed a similar pattern of results, using cluster analysis. They identified a group of college students who showed low levels of attachment to parents and

high levels of emotional autonomy, but scored high on measures of psychosocial adjustment. The authors concluded that these individuals were able to emerge from what can be considered a less healthy family system, with healthy psychological adjustment, thanks to high levels of emotional autonomy. Conclusions of both of these studies are in sharp contrast with the results of Lamborn and Steinberg (1993) and Delaney (1996).

These differing conclusions could be partly explained by the fact that different versions of the EAS were used in these studies. Furthermore, Fuhrman and Holmbeck's (1995) small sample comprised a majority of African-Americans, and they represented a broad age range (10–18 years). On the other hand, Lamborn and Steinberg's very large sample was mainly composed of Caucasian Americans from a narrower age range (14–18 years) (Lamborn and Steinberg, 1993). Tentatively, one could also argue that the differing conclusions have to do with the methods of analysis used. Lamborn and Steinberg (1993) compared mean levels of adjustment in groups of adolescents with different levels of autonomy and parental support using ANOVA and *t*-tests. Fuhrman and Holmbeck (1995) on the contrary, tested true moderator effects using multiple regression in which an interaction term (family context $\times$ emotional autonomy) was entered. However, taking into account these possible explanations, questions remain about the precise nature of emotional autonomy and its developmental outcomes in adolescence.

One way to clarify the nature of emotional autonomy in adolescence is to compare it with another important aspect of autonomy development in early adolescence, namely behavioural autonomy (Hill and Holmbeck, 1986; Steinberg, 1995). Behavioural autonomy is the capacity to make independent decisions regarding all kinds of behaviours (Bosma *et al.*, 1996) and therefore is sometimes called decisional autonomy. Unlike emotional autonomy, behavioural autonomy in early adolescence is consistently associated with psychosocial adjustment. Many studies (e.g. Lamborn *et al.*, 1996) have shown that behavioural autonomy in early adolescence predicts greater deviance and lower academic competence. In the present study, we compared emotional and behavioural autonomy as predictors of psychosocial adjustment. Some aspects of adjustment are probably related specifically to emotional autonomy, while others are related to behavioural autonomy. To know which relates to which, both aspects of autonomy must be studied together. This was done using path analysis to predict psychosocial adjustment from autonomy and adding a path between emotional and behavioural autonomy, thus controlling for the positive correlation between the two aspects of autonomy.

As mentioned by both Lamborn and Steinberg (1993) and Fuhrman and Holmbeck (1995), it is useful to study the association between adolescent autonomy and psychosocial adjustment in the context of the broader relationship with parents. Rather than looking at a single aspect of the relationship with parents (e.g. support), we described the relationship with parents in terms of the parenting process as perceived by the adolescent (Darling and Steinberg, 1993). A broad picture of the quality of parenting as perceived by the adolescent was constructed by measuring three well-known dimensions of parenting. Many studies indicated that a pattern of high levels of both warmth/involvement and monitoring/parental knowledge, combined with low levels of psychological control (called authoritative parenting), is most adaptive for young adolescents' psychosocial adjustment (for reviews, see Baumrind, 1989, 1991). In this study we expect that being autonomous relates differently to adolescent adjustment for adolescents experiencing different qualities of the relationship with parents. This hypothesis was tested in three different ways. First, we directly assessed Lamborn and Steinberg's (1993) hypothesis that patterns of adjustment differ according to

levels of parenting quality and autonomy, by studying how these variables interact in their associations with mean levels of adjustment. Second, as Fuhrman and Holmbeck (1995) did, we studied possible moderating effects of parenting quality in the prediction of adjustment by emotional autonomy, by using multiplicative terms in multiple regression models. Finally, we examined path models predicting adjustment by autonomy. In these models parenting was partialled out by introducing a path from parenting quality to autonomy. Autonomy therefore mediated the relationship between the parenting context and the adjustment variables in the latter analyses.

Some authors have suggested that age (Frank *et al.*, 1990; Lamborn and Steinberg, 1993) and gender (Steinberg and Silverberg, 1986) might also play a role in understanding the mixed outcomes associated with emotional autonomy. In the present study, age and gender were studied in an exploratory way as possible moderators in the relation between autonomy and psychosocial adjustment. Emotional autonomy might be a less stressful experience for older adolescents, especially males.

Methods

Participants

The sample comprised all students ($n = 558$) from the academic track from two secondary schools in the province of Antwerp, in the Dutch-speaking part of Belgium. These schools mainly attract middle-class, Caucasian students from rural areas. A breakdown by grade and gender yielded 244 participants from Grade 7 (135 girls, 109 boys), 165 participants from Grade 9 (94 girls, 71 boys), and 149 participants from Grade 11 (85 girls, 64 boys). Mean ages for these groups were 12 years and 10 months ($S.D. = 3.9$ months), 14 years and 10 months ($S.D. = 4.5$ months), and 16 years and 10 months ($S.D. = 4.3$ months). A large majority of these adolescents (92%) came from intact families.

Letters describing the study were sent to the parents of all students from the three grades, asking for passive consent. Only 20 students (3%) did not have their parents' permission to participate.

Measures

Most of the instruments used were initially developed in English and were translated into Dutch for the present study. Both authors made independent translations of these scales and agreed upon a common version, following discussion. This preliminary version was then double-checked by a third person with a Ph.D. in Developmental Psychology.

Emotional autonomy. Emotional autonomy from parents was measured using a shortened (12-item) version of Steinberg and Silverberg's (1986) Emotional Autonomy Scale (EAS). Following Lamborn and Steinberg (1993), the items referring to perceptions of parents as individuals beyond the parental role were dropped from the scale in the present study. Previous work indicated that this aspect of emotional autonomy is very slow to develop during adolescence (Steinberg and Silverberg, 1986) and these items also had very low and even negative item-total correlations. Two additional items from the original Individuation subscale were dropped because of very low item-total correlations. These items were "My parents would be surprised to know what I'm like when I'm not with them" and "I wish my parents would understand who I really am". The internal consistency of this shortened

version was 0.82 for this sample. (The internal consistency of the 20-item version, as reported by Steinberg and Silverberg, was 0.75.) Items were rated on a four-point Likert scale ranging from "strongly agree" to "strongly disagree". A sample item, reverse scored, was "My parents hardly ever make mistakes".

Behavioural autonomy. The Perspectives on Adolescent Decision Making Questionnaire B (PADM-B; Bosma *et al.*, 1996) is a 21-item measure in which participants are asked to use a five-point scale to indicate, issue by issue, the extent to which the adolescent or the parent decides what should happen. Sample issues are "whether to smoke or not", or "who to go around with". Higher levels of behavioural or decisional autonomy often lead to conflict between adolescents and their parents (Bosma *et al.*, 1996) and predict greater deviance and lower academic competence (Lamborn *et al.*, 1996). The PADM-B showed good reliability, as defined by Cronbach's alpha (0.87).

Authoritativeness. To present a broad picture of parenting quality as perceived by the adolescent, we used the Parenting Scales (Lamborn *et al.*, 1991; Sessa, 1992). The instrument consists of three scales, each measuring a well-known dimension of parenting. For this study the original instrument was slightly altered, in that students were required to answer all items on a five-point Likert scale ranging from "strongly disagree" to "strongly agree". Involvement (11 items), an indicator of high parenting quality, measures the extent to which the adolescent perceives his or her parents as loving, responsive and involved. The scale showed good reliability (Cronbach's alpha = 0.81). A sample item is "I can count on my father to help me out, if I have some kind of problem". Three items of the original monitoring subscale were dropped. The items that were dropped were those that ask how much the parents are *trying* to get to know where the adolescent is and what he or she does when not at home. This resulted in a six-item scale with acceptable reliability (Cronbach's alpha was 0.70, as compared with an alpha of 0.52 for the full nine-item version of the scale). This scale does not really measure monitoring (i.e. active tracking and surveillance by parents), but rather measures the amount of knowledge parents have about their teenagers' activities. A sample item is "My parents know exactly where I am most of the time after school". In a study of Swedish adolescents, Stattin and Kerr (in press) showed that parental knowledge does not come from active tracking by parents, but rather from spontaneous self-disclosure by the adolescent. Therefore monitoring was questioned as a measure of parenting behaviour. However, the same authors suggested that in families where parental knowledge is high, the parents might have done something to build up the kind of relationships that facilitate communication. Parental knowledge in this sense continues to be an indicator of high parenting quality. The psychological control subscale (12 items) describes covert, psychological methods of controlling the adolescent's activities and behaviours, such as guilt induction, love withdrawal and excessive pressure for change (Barber, 1996). Unlike parental knowledge, psychological control has a negative effect on adolescent development (Steinberg, 1990) and therefore was taken as an indicator of poor parenting quality. The scale showed acceptable reliability (Cronbach's alpha = 0.74). A sample item is "When I get a poor grade at school, my parents make me feel guilty".

Taking these three scales together to get a broad picture of parenting quality is acceptable, since the three subscales (with psychological control reverse scored) show good internal consistency (Cronbach's alpha = 0.55 for three "items"). An additional check using exploratory factor analysis (principal components method) revealed that a single factor solution for the three dimensions of parenting explains 53% of the variance in these

variables. Therefore, and because authoritative parenting is shown to be the most efficient style of parenting in early and middle adolescence, a continuous measure of authoritativeness was constructed in this study, rather than the categorical measures created by other authors (Steinberg *et al.*, 1991, 1992). A weighted sum (weighted by the number of items in each scale) of involvement, monitoring and low psychological control was computed for each participant. High scores on this variable indicate high-quality, or authoritative, parenting.

Self-reliance. The self-reliance subscale of the Psychosocial Maturity Inventory (Form D) (Greenberger *et al.*, 1974) measures initiative, an absence of excessive dependency on others and a sense of control over one's life. The scale is composed of 10 items and showed adequate internal consistency in this study (Cronbach's $\alpha = 0.70$). Sample items, reverse scored, are "Luck decides most things that happen to me" and "In a group I prefer to let other people make the decisions". Students answered all items on five-point Likert scales ranging from "strongly disagree" to "strongly agree".

Internal distress. The measure used for this variable is the Internal Distress Scale (IDS; Weinmann, 1992) which is composed of 13 items derived from the Depression Scale of the Center for Epidemiological Studies (CES-D; Radloff, 1977). Items refer to somatic and psychological symptoms of internal distress, such as headaches, stomach-aches, colds, anxiety, tension and depression. Participants were asked to indicate how often they experienced these symptoms during the last month. Cronbach's α was 0.77.

School grades. This measure consisted of self-reported grade point average for a semester, expressed as a percentage. The validity of this measure is partially supported by the non-significant correlation with social desirability ($r = 0.07$).

Deviant behaviour. The Deviant Behaviour Scale (DBS; Weinmann, 1992) is a 13-item scale consisting of various deviant or delinquent behaviours that differ in severity (e.g. using marijuana or using hard drugs). The deviant behaviours include drug use, school misconduct and antisocial behaviour. Subjects were asked to indicate how often they had engaged in these behaviours during the last 6 months. This scale was used in Grades 9 and 11 only. Cronbach's α was 0.71 in this subsample.

Social desirability. Some of the self-report measures used in this study were influenced by social desirability. The positive correlation between behavioural autonomy and internal distress ($r = 0.15$, $p < 0.01$), for instance, became non-significant when social desirability was partialled out ($r = 0.07$). All analyses in this study were therefore controlled for social desirability. As the measure of social desirability, a shortened Dutch adaptation (13 items; Denayer, 1974) of the original scale by Crowne and Marlowe (1960) was used. This version was specifically developed for use with adolescents. Cronbach's α was 0.70.

Procedure

The participants completed all the measures in group sessions during regularly scheduled classes. The first author supervised all of these test sessions, which took no longer than 50 min. Following a brief introduction about the purpose of the study, the instructions for the entire questionnaire were read aloud to the participants and any questions regarding the way in which the measures should be completed, were answered.

Results

Correlational analysis

Pearson product-moment correlations between authoritativeness, emotional and behavioural autonomy and different aspects of psychosocial adjustment, as well as correlations between all these variables and social desirability, are included in Table 1. High levels of authoritativeness are associated with lower levels of autonomy, especially emotional autonomy. As expected, authoritativeness is also associated with patterns of good psychosocial adjustment (greater self-reliance, less internal distress, better school grades, and lower levels of deviant behaviour). Finally, authoritativeness seems to be highly influenced by social desirability, which corroborates our initial option to control all analyses for this biasing factor.

Emotional and behavioural autonomy are highly correlated. These two aspects of autonomy show the same patterns of correlations with the psychosocial adjustment variables. Higher levels of autonomy are associated with greater self-reliance, more internal distress, lower school grades and more deviant behaviour. The size of the correlations, however, differs for these two aspects of autonomy. While emotional autonomy is more strongly associated with internal distress [$t(555) = 3.75$; $p < 0.001$], behavioural autonomy is more strongly associated with self-reliance [$t(555) = 3.66$; $p < 0.001$]. Both dimensions of autonomy seem to be undesirable, as shown by their negative correlations with social desirability. Correlations between the psychosocial adjustment variables are as expected. Somewhat unexpectedly, self-reliance fails to show a significant correlation with the other adjustment variables or with social desirability.

Age and gender differences

A $3(\text{grade}) \times 2(\text{gender})$ MANCOVA, with authoritativeness, emotional autonomy, behavioural autonomy, self-reliance, internal distress and school grades as dependent variables and social desirability as a covariate, yielded significant effects (using Wilk's lambda) for grade [$F(12,1092) = 30.61$; $p < 0.001$], gender [$F(6,546) = 17.97$; $p < 0.001$], and the interaction between them [$F(12,1092) = 3.15$; $p < 0.001$]. Subsequent univariate analyses were performed. The analysis with deviant behaviour as the dependent variable consisted of a $2(\text{grade}) \times 2(\text{gender})$ ANCOVA, as deviant behaviour was only measured in Grades 9 and 11. Adjusted mean scores on all variables as a function of grade and gender are presented in Table 2. The analyses revealed significant gender differences for most variables. Girls

Table 1 *Pearson correlations among all variables*

Variable	2	3	4	5	6	7	8
1. Authoritativeness	-0.56**	-0.31**	0.12**	-0.26**	0.26**	-0.50**	0.42**
2. Emotional autonomy		0.43**	0.17**	0.31**	-0.19**	0.28**	-0.37**
3. Behavioural autonomy			0.33**	0.15**	-0.18**	0.33**	-0.26**
4. Self-reliance				0.00	-0.03	0.11	-0.01
5. Internal distress					-0.13**	0.19**	-0.34**
6. School grades						-0.41**	0.07
7. Deviant behaviour							-0.22**
8. Social desirability							

$N = 558$ in all cases, except for deviant behaviour, where $n = 314$.

* $p < 0.05$; ** $p < 0.01$.

reported higher levels of authoritativeness [$F(1,551) = 10.05$; $p < 0.01$], emotional autonomy [$F(1,551) = 16.81$; $p < 0.001$], and internal distress [$F(1,551) = 50.13$; $p < 0.001$], but lower levels of deviant behaviour [$F(1,309) = 18.56$; $p < 0.001$], than boys.

The ANCOVAs also revealed consistent age differences for all variables, except for internal distress. Emotional autonomy [$F(2,551) = 41.96$; $p < 0.001$], behavioural autonomy [$F(2,551) = 156.70$; $p < 0.001$], self-reliance [$F(2,551) = 37.63$; $p < 0.001$] and deviant behaviour, [$F(1,309) = 26.23$; $p < 0.001$] increased linearly over the age range studied. Students in the higher grades reported lower levels of authoritative parenting [$F(2,551) = 14.38$; $p < 0.001$] and poorer school grades [$F(2,551) = 23.46$; $p < 0.001$], than students in the lower grades. In line with the Steinberg and Silverberg (1986) study, the age difference for emotional autonomy between Grades 7 and 9 was far bigger than that between Grades 9 and 11.

In addition, the grade $\times$ gender interaction was significant for behavioural autonomy, internal distress and school grades [$F(2,551) = 3.14$; $p < 0.05$, 5.37 ; $p < 0.01$, and 3.98 ; $p < 0.05$, respectively]. Inspection of cell means indicated that the level of behavioural autonomy in Grade 7 was much higher for girls than it was for boys, whereas it was very similar in Grades 9 and 11. While boys did not show any age differences in internal distress, girls showed a steep increase in distress between Grades 7 and 9. For girls, the main decrease in school grades took place between Grades 9 and 11, whereas for boys this occurred between Grades 7 and 9.

Mean adjustment as a function of authoritativeness and autonomy and their interaction

To assess directly Lamborn and Steinberg's (1993) hypothesis that being autonomous relates differently to adjustment for adolescents with qualitatively different parent-child relationships, we performed this set of analyses. A median split on the authoritativeness, emotional autonomy and behavioural autonomy scores divided participants into high and low groupings on these variables. Next, a series of $2(\text{authoritativeness}) \times 2(\text{autonomy}) \times 3(\text{grade}) \times 2(\text{gender})$ ANCOVAs was performed with the psychosocial adjustment variables as dependent variables and social desirability as a covariate. Testing the above hypothesis, we looked especially at the interaction between authoritativeness and autonomy. For the

Table 2 *Adjusted mean scores on all variables as a function of grade and gender, controlled for social desirability*

Variable	Grade			Gender	
	7	9	11	F	M
Authoritativeness	5.82 ^a	5.49 ^b	5.08 ^c	5.64	5.29
Emotional autonomy	31.51 ^a	34.23 ^b	35.78 ^c	34.63	33.05
Behavioural autonomy	67.93 ^a	75.02 ^b	85.22 ^c	76.80	75.31
Self-reliance	34.67 ^a	36.37 ^b	39.13 ^c	37.06	36.39
Internal distress	9.25 ^a	10.51 ^a	9.81 ^a	11.45	8.26
School grades	73.24 ^a	70.70 ^b	67.11 ^c	71.01	69.69
Deviant behaviour	NA	5.55 ^a	7.65 ^b	5.71	7.49

For grade, means in the same row that do not share superscripts differ at $p < 0.05$.

NA, not applicable.

adjustment variables that were measured in all age groups, these were preceded by a MANCOVA. Separate analyses were run with emotional and behavioural autonomy as independent variables. Effects of grade and gender, which were entered as control variables, were very similar to those reported in Table 2.

The MANCOVA, with emotional autonomy as an independent variable and the adjustment measures as dependent variables, revealed significant multivariate main effects (using Wilks' lambda) for both authoritativeness [$F(3,531) = 11.82$; $p < 0.001$] and emotional autonomy [$F(3,531) = 3.30$; $p < 0.05$]. However, the interaction between these two variables did not reach significance [$F(3,531) = 0.59$; $p > 0.60$]. Subsequent univariate analyses revealed that adolescents experiencing high levels of authoritativeness in the relationship with parents showed higher levels of self-reliance [$F(1,533) = 14.09$; $p < 0.001$], lower levels of internal distress [$F(1,533) = 14.44$; $p < 0.001$], better school grades [$F(1,533) = 10.45$; $p < 0.01$] and lower levels of deviant behaviour [$F(1,297) = 23.29$; $p < 0.001$] than those experiencing low levels of authoritativeness. Effects for emotional autonomy were far less substantial. Adolescents reporting high levels of emotional autonomy showed slightly higher levels of self-reliance [$F(1,533) = 3.98$; $p < 0.05$], did a little worse in school [$F(1,533) = 5.13$; $p < 0.05$] and engaged in slightly more deviant behaviour [$F(1,297) = 4.87$; $p < 0.05$]. In this analysis, the variance in internal distress seemed to be almost completely accounted for by the effects of authoritativeness and gender. Therefore, no effect of emotional autonomy on internal distress was found.

The MANCOVA with behavioural autonomy as an independent variable showed very much the same pattern of results. Again, significant multivariate main effects were found for both authoritativeness [$F(3,532) = 6.52$; $p < 0.001$] and behavioural autonomy [$F(3,532) = 5.32$; $p < 0.01$] on the adjustment variables. However, the interaction between these two variables did not reach significance [$F(3,532) = 2.58$; $p > 0.05$]. (Notice that in this analysis there was one degree of freedom more in the error term than in the previous one. This was due to one cell in the design-matrix that was empty.) Subsequent univariate analyses revealed that adolescents who experienced high levels of authoritativeness in the relationship with parents showed lower levels of internal distress [$F(1,534) = 11.39$; $p < 0.01$], better school grades [$F(1,534) = 6.83$; $p < 0.01$] and lower levels of deviant behaviour [$F(1,298) = 16.21$; $p < 0.001$] than those who experienced low levels of authoritativeness. Adolescents who reported high levels of behavioural autonomy only showed higher levels of self-reliance [$F(1,534) = 14.04$; $p < 0.001$] than those who reported low levels of behavioural autonomy.

A direct test of Lamborn and Steinberg's (1993) assumption of a significant interaction between the parenting context and emotional autonomy in relation to mean levels of adjustment, therefore led us to reject this hypothesis. Less stringent tests, as performed by Lamborn and Steinberg (i.e. a comparison of mean levels of adjustment in four groups, defined by combining high and low scores on both authoritativeness and autonomy), were thus deemed unnecessary.

Predicting adjustment scores from authoritativeness and autonomy and their interaction

The next set of analyses was performed to examine the possibility that authoritativeness might moderate the relation between adjustment and autonomy. A moderator effect means that the nature of the relationship between autonomy and psychosocial adjustment varies, depending on the amount of authoritativeness the adolescent experiences in the family

(Jaccard *et al.*, 1990). This analysis was modelled following the work of Fuhrman and Holmbeck (1995). A set of hierarchical multiple regression analyses was applied with each of the four adjustment variables as dependent variable. Control variables (grade, gender and social desirability) were entered in Step 1 of the analysis, the main effects (authoritativeness and autonomy) in Step 2 and the interaction (authoritativeness $\times$ autonomy) in Step 3. Separate analyses were run for emotional and behavioural autonomy. Raw scores for authoritativeness and autonomy were centred for these analyses (Cronbach, 1987), in order to reduce the problem of multicollinearity (very high correlations between the interaction term and the independent variables).

We can be very brief about the results of these analyses. The main effects largely confirmed the pattern of correlations presented in Table 1. The results of the interaction term in all models confirmed the previous results. Adding the interaction term produced a barely significant R^2 change ($p < 0.05$) in only one of the eight models. This was the case in the model predicting self-reliance by authoritativeness and emotional autonomy. Subsequent regression analyses revealed that emotional autonomy predicted significantly higher levels of self-reliance in the group of adolescents who reported low levels of authoritativeness ($\beta = 0.13$; $p < 0.05$) but not in those who reported high levels of authoritativeness ($\beta = 0.10$; $p > 0.10$). Given this single, very small interaction, we have to reject the overall hypothesis that there is an interaction between parenting quality and autonomy in the prediction of psychosocial adjustment in adolescence.

Path models testing the relations between authoritativeness, autonomy and adjustment

Given the non-significant interaction between parenting quality and autonomy in predicting or comparing levels of adjustment, we examined the main effects of parenting quality and autonomy more clearly, using path analysis. Path analysis was used to derive the best-fitting model among all the variables in this study. The system of equations in this model can be split up into two sets of equations. We have very specific *a priori* hypotheses for the relationships among the independent variables (authoritativeness, emotional and behavioural autonomy). Authoritativeness is believed to be the context for autonomy development (Darling and Steinberg, 1993). Therefore it precedes the autonomy variables in the path model. Furthermore, age differences in autonomy indicated that emotional autonomy mainly increased during the early part of adolescence, whereas behavioural autonomy developed gradually during the whole of adolescence. Only variables preceding other variables can have causal effects on the latter, so we assume a path from emotional autonomy to behavioural autonomy. Given the high correlation between emotional and behavioural autonomy, it is hypothesized that the indirect effect of authoritativeness on behavioural autonomy (through emotional autonomy) accounts for the total correlation between these variables. The theoretical model based on these assumptions is presented in Figure 1.

Based on the correlations in Table 1, we can also make some exploratory assumptions for the relations among the independent variables and the psychosocial adjustment variables. We can, for instance, assume a negative path from authoritativeness to deviant behaviour. As the relationships among the psychosocial adjustment variables were not the scope of this study, all analyses were performed separately for each adjustment variable.

Analyses were performed using the LISREL 8 package, using the Simplis command language (Jöreskog and Sörbom, 1993). The input for all analyses was a matrix of covariances. A double rule-of-thumb was used to accept a final model. First, all parameters in

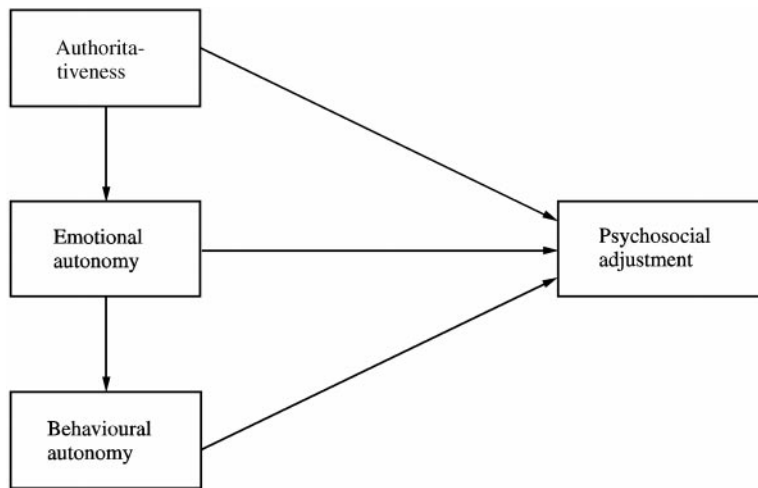


Figure 1. Theoretical path-analytic model. Influence of authoritativeness, emotional autonomy and behavioural autonomy on psychosocial adjustment.

the model had to be significant ($p < 0.05$). Second, none of the residual covariances could be significant. This approach eventually resulted in a model with acceptable overall fit, as indicated by χ^2 . Social desirability was entered as a control variable in all analyses, by specifying paths between social desirability and all other variables in the model. As a result, social desirability did not influence the overall fit of the model. Although age (or grade) was associated with almost all variables in this study, it was not entered as an additional control variable because of its categorical nature.

Initially, the subpart of the model that described the relations between authoritativeness, emotional autonomy and behavioural autonomy (the left hand side of Figure 1) was fitted to the data. This model, with 1 df (the path from authoritativeness to behavioural autonomy was fixed to zero), fitted the data well in the total sample ($\chi^2 = 2.43$; $p > 0.10$). Authoritativeness thus predicted lower levels of emotional autonomy, which in turn caused higher levels of behavioural autonomy. A direct test of both grade and gender differences, using a multigroup analysis, proved that this model differs significantly between boys and girls, and between Grades 7, 9 and 11. Therefore, following model testing in the total sample, separate analyses were run for these different groups.

Table 3 presents standardized path coefficients of the full model in the total sample and different subsamples. The paths describing the effects of social desirability were not included; neither was the path from authoritativeness to behavioural autonomy that was fixed to zero in all analyses. In the total sample, all three independent variables had positive significant effects on self-reliance. The equation for self-reliance was, therefore, unrestricted and did not influence the overall fit. Although the direct effect of emotional autonomy on self-reliance is much smaller than the effect of behavioural autonomy on self-reliance, the total effects are very similar (0.35 and 0.34, respectively). The total effect of emotional autonomy on self-reliance consists of both direct and indirect effects over behavioural autonomy. In the same vein, authoritativeness had both a positive direct effect and a smaller, but negative, indirect effect on self-reliance.

Table 3 Standardized path coefficients describing the relationships between authoritative (A), emotional autonomy (E), behavioural autonomy (B) and psychosocial adjustment (P)

Adjustment variable (P)	Path					df	χ^2
	A-E	E-B	A-P	E-P	B-P		
Total sample ^a							
Self-reliance	-0.49**	0.39**	0.34**	0.22**	0.34**	1	2.43
Internal distress	-0.49**	0.39**	0	0.22**	0	3	3.75
School grades	-0.49**	0.39**	0.25**	0	-0.12**	2	2.96
Deviant behaviour	-0.49**	0.20**	-0.43**	0	0.24**	2	4.28
Girls ^b							
Self-reliance	-0.51**	0.28**	0.35**	0.23**	0.28**	1	1.48
Internal distress	-0.51**	0.28**	0	0.25**	0	3	5.43
School grades	-0.51**	0.28**	0	-0.22**	0	3	6.94
Deviant behaviour	-0.54**	0	-0.41**	0	0.18**	2	0.89
Boys ^c							
Self-reliance	-0.55**	0.50**	0.33**	0.20**	0.41**	1	2.50
Internal distress	-0.55**	0.50**	-0.16**	0	0	3	2.59
School grades	-0.55**	0.50**	0.39**	0	0	3	5.76
Deviant behaviour	-0.53**	0.31**	-0.37**	0	0.33**	2	1.11
Grade 7 ^d							
Self-reliance	-0.46**	0.34**	0.30**	0.23**	0.13*	1	0.02
Internal distress	-0.46**	0.34**	0	0.22**	0	3	2.36
School grades	-0.46**	0.34**	0.17*	0	0	3	0.09
Grade 9 ^e							
Self-reliance	-0.47**	0	0.33**	0	0.23**	3	5.16
Internal distress	-0.47**	0	0	0.25**	0	4	3.25
School grades	-0.47**	0	0.25**	0	0	4	5.80
Deviant behaviour	-0.47**	0	-0.47**	0	0.13*	3	4.89
Grade 11 ^f							
Self-reliance	-0.49**	0.18*	0.38**	0.18*	0.20*	1	0.80
Internal distress	-0.49**	0.18*	0	0	0	4	2.62
School grades	-0.49**	0.18*	0.22**	0	0	3	2.17
Deviant behaviour	-0.49**	0.18*	-0.38**	0	0.20**	2	1.51

Non-significant paths were fixed to zero.

^aN = 558 in all models, except for deviant behaviour, where n = 314.

^bn = 314 in all models, except for deviant behaviour, where n = 179.

^cn = 244 in all models, except for deviant behaviour, where n = 135.

^dn = 244.

^en = 165.

^fn = 149.

*p < 0.05; **p < 0.01.

The pattern of results for the other adjustment variables clarified the difference between emotional and behavioural autonomy. Emotional autonomy was the only independent variable that had a direct effect on internal distress. However, it did not have a direct effect on school grades or deviant behaviour. The latter variables were predicted directly by both authoritative and behavioural autonomy.

Separate analyses by gender indicated that the association between emotional and behavioural autonomy is much stronger in boys than in girls. The pattern of results for self-reliance was very similar, but not identical, for boys and girls. Among boys, behavioural autonomy had a larger effect (both direct and total effect) on self-reliance than it had among girls. Emotional autonomy was not a stressful experience for boys, whereas it was for girls. In boys this was compensated by the fact that authoritativeness had a negative effect on internal distress. The same difference between the genders held for school results. Emotional autonomy in boys did not have an effect on school grades, but authoritativeness did. The fact that behavioural autonomy no longer predicted school grades in either of the two subsamples may be due to the smaller sample size, as compared to the total sample. The pattern of results for deviant behaviour was similar for boys and girls. In this analysis, however, the effect of emotional on behavioural autonomy was no longer significant in the subsample of girls. A significant direct path from authoritativeness to behavioural autonomy (not included in Table 3) compensated for this. This pattern of results, however, did not result from the deviant behaviour in itself, but rather from the smaller, older sample used for analysis.

Separate analyses by grade revealed that the effect of emotional autonomy on behavioural autonomy is much smaller in the higher grades than it is in Grade 7. For Grade 9 this effect (0.09) fell below significance and was therefore fixed to zero. Also in Grade 9, emotional autonomy did not have an effect on self-reliance. Emotional autonomy led to higher levels of internal distress in Grades 7 and 9, but not in Grade 11. Authoritativeness predicted better school grades at all grade levels. While authoritativeness was associated with lower levels of deviant behaviour, especially in Grade 9, behavioural autonomy led to more deviant behaviour, especially in Grade 11. This may be due to the higher levels of authoritativeness and behavioural autonomy in Grades 9 and 11, respectively (see Table 2).

In summary, then, introducing both authoritativeness (parenting quality) and different aspects of autonomy into our explanatory model appeared useful. Partialling out the effect of parenting quality clarified the importance of parenting context on the one hand and autonomy on the other, and revealed the differential effects of emotional and behavioural autonomy in the prediction of psychosocial adjustment in adolescence.

Discussion

Pearson correlations between all variables in this study support Ryan and Lynch's (1989) conclusion that adolescents who detach themselves from their parents, as indicated by high scores on the measure of emotional autonomy, are at risk for unhealthy psychosocial development. However, an important exception is the consistent positive association between emotional autonomy and self-reliance, even when controlled for social desirability. This positive association is especially remarkable as Steinberg and Silverberg (1986) reported a curious and unexpected, but consistently negative, association between emotional autonomy and feelings of self-reliance. The main reason for this contradiction might be the shortened 12-item scale used to measure emotional autonomy in this study, as opposed to the full 20-item version. In earlier research, Frank *et al.* (1990) concluded that the items that were dropped from the scale in this study had a pejorative and, in some instances, somewhat paranoid tone that suggested alienation and distrust.

Applying statistically sound models to test directly the hypothesis that an interaction exists between parenting quality and autonomy, both when comparing mean levels of adjustment (ANOVA) and predicting adjustment (multiple regression), leads in both cases to an overall rejection of this hypothesis. On the one hand it appears that mean levels of adjustment do differ as a function of both authoritativeness and autonomy, but that these main effects do not differ according to the level of the other independent variable. Main effects of authoritativeness and autonomy in these ANOVA models clearly supported the results of simple correlations. On the other hand, the rejection of the above hypothesis means that the slope in the regression of adjustment on autonomy does not differ as a function of the moderator variable, in this case authoritativeness. However, one must add here that the tradition of accepting the baseline model (without interaction) when statistical significance is not found has undoubtedly created difficulty in bringing out interaction effects (Cronbach, 1987). A meta-analysis of simulated data has shown that even with large sample sizes, moderator effects of potential interest are likely to be called non-significant (Sackett *et al.*, 1986).

One exception to this general rejection of the interaction hypothesis is the slightly significant interaction found in the model predicting self-reliance by authoritativeness and emotional autonomy. The fact that an interaction was found in this regression analysis and not in the ANOVA may be due to the continuous character of the variables included in the regression, compared to the dichotomous variables in the ANOVA. Non-significant higher-order interactions between control variables (grade and gender) and independent variables (authoritativeness and autonomy) included in the full factorial model might also explain this small difference. Interpretation of the above interaction confirms Fuhrman and Holmbeck's (1995) finding that when family cohesiveness (parenting quality) was low, emotional autonomy predicted higher levels of adolescent competence (self-reliance). When cohesiveness (parenting quality) was high, however, emotional autonomy was associated with less competence (self-reliance).

Given the non-significant interaction between parenting quality and autonomy when predicting or comparing levels of adjustment, we examined the main effects of parenting quality and autonomy using path analysis. By introducing both independent variables into a single path model we accounted for the simultaneous effects of both autonomy and the family context on psychosocial adjustment. In addition to the significant association with self-reliance, we saw in the total sample that emotional autonomy predicted higher levels of internal distress only (an internalized aspect of adjustment). The previously reported association between emotional autonomy and the behavioural aspects of psychosocial adjustment, such as school grades and deviant behaviour (Turner *et al.*, 1991, 1993; Fuhrman and Holmbeck, 1995; Chen and Dornbusch, 1998), seems spurious when behavioural autonomy is taken into account. This finding confirms our view that autonomy during adolescence is a multidimensional process. The positive but moderate correlation between emotional and behavioural autonomy further confirms this multidimensional structure of autonomy. Silverberg and Gondoli's (1996) suggestion that the subtle yet important distinctions between emotional autonomy and other aspects of autonomy that are associated more consistently with psychosocial adjustment should be borne in mind, therefore, seems to be valid.

As authoritativeness was entered first in the path models tested, all previous conclusions are valid only when taking authoritativeness (the broader family context) into account. The way in which this was done in this study, i.e. by creating a weighted sum of

involvement, monitoring and low psychological control which resulted in a continuous measure of authoritativeness, is only one way of looking at the quality of parenting and it seems to cause some problems. In the total sample, for instance, authoritativeness has both a positive direct effect and a negative indirect effect on self-reliance. This raises questions about what it means to experience authoritative, or more specifically non-authoritative, parenting. A more differentiated way of looking at the quality of parenting is to define qualitatively different parenting style groups, as outlined by Baumrind (1971) and Maccoby and Martin (1983). Indeed, experiencing non-authoritative parenting can have different meanings, such as being rejected by one's parents, having authoritarian parents or having permissive parents.

Becoming autonomous is a more stressful experience for girls than it is for boys. This finding confirms Lamborn and Steinberg's (1993) finding. However, if we take biological age into account, the effect might be spurious. Girls also show higher levels of emotional autonomy than boys (Steinberg and Silverberg, 1986). Steinberg (1987) showed that pubertal maturation has a strong impact on the emotional separation between adolescents and their parents. Pubertal maturation therefore accelerates the process of autonomy development. When we take into account the well-established fact that girls are about 2 years ahead of boys in their levels of biological maturation during early and middle adolescence, it is no wonder that girls show higher levels of emotional autonomy. High levels of emotional autonomy indeed seem to be associated with a certain amount of internal distress, as has already been mentioned by Blos (1979).

However, in Grade 11, where the levels of emotional autonomy were higher than those in Grades 7 and 9, this aspect of autonomy did not predict internal distress. It seems that adolescents who are in the middle of the process of emotional autonomy development experience this as rather stressful. Once emotional autonomy becomes a more established aspect of the adolescent's identity, it is no longer associated with internal distress. This view confirms the results of Frank *et al.* (1990), who suggested that it is not until late adolescence that emotional autonomy predicts positive profiles of adjustment.

The term "structural" in structural equations modelling stands for the assumption that the parameters in the path models are not just descriptive measures of association but rather that they reveal an invariant "causal" relation (Bollen, 1989). However, the model presented in this study is a static model, meaning that all variables, both independent and dependent, were measured at the same point in time. This makes it very hard to make real causal inferences, especially for the path model. Given the stability of variables, the temporal order presented (e.g. emotional autonomy preceding behavioural autonomy) is a very weak approximation of directed influences. A follow-up of the same participants in a longitudinal study, which is currently in progress, will permit us to make causal inferences concerning the relationships between autonomy, parenting and psychosocial adjustment. Furthermore, it will permit us to confirm the growth curves of the different variables in the study which were clearly suggested by the reported age differences.

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